Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

If $|z| \geq 5$, then least value of $\left|z - \frac{1}{z}\right|$ is

(A) 5

(B) $24/5$

(C) 8

(D) $8/3$

Q.2

For all complex numbers z_1, z_2 satisfying $|z_1| = 12$ and $|z_2 - 3 - 4i| = 5$, the minimum value of $|z_1 - z_2$ is

(A) 0

(B) 2

(C) 7

(D) 17

Q.3

If $\bar{z} = 3i + \frac{25}{z+3i}$, then $|z|$ cannot exceed

(A) 3

(B) 8

(C) 16

(D) 18

Q.4

The inequality $|z - 4| < |z - 2|$ represents the region given by

(A) $\mathrm{Re}(z) \geq 0$

(B) $\mathrm{Re}(z) < 3$

(C) $\mathrm{Re}(z) \leq 0$

(D) $\mathrm{Re}\, z > 3$

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Q.5

If $\Delta = \begin{vmatrix} 6i & -3i & 1 \\ 4 & 3i & -1 \\ 20 & 3 & i \end{vmatrix} = x + iy$, then

(A) $x = 3, y = 1$

(B) $x = 1, y = 3$

(C) $x = 0, y = 3$

(D) $x = 0, y = 0$

Q.6

If z is a complex number satisfying $Rez + Imz = 4$, then z cannot be

(A) $\sqrt{\frac{17}{2}}$

(B) $\sqrt{10}$

(C) $\sqrt{7}$

(D) $\sqrt{8}$

Q.7

If z_1, z_2 and z_3 are 3 distinct complex numbers such that $\frac{3}{z_1 - z_2} = \frac{5}{z_2 - z_3} = \frac{7}{z_3 - z_1}$, then the value of $\frac{9}{z_1 - z_2} + \frac{25}{z_2 - z_3} + \frac{49}{z_3 - z_1}$ is equal to

(A) 0

(B) 1

(C) -1

(D) 15

Q.8

Let $O = 0,0$, $A = 3,0$, $B = 0, -1$ and $C = 3,2$, then the minimum value of $z + z - 3 + z + i + z - 3 - 2i$ occurs at the (where, z is a complex number)

(A) point of intersection of AB and CO

(B) point of intersection of AC and BO

(C) point of intersection of CB and AO

(D) Mean of O, A, B, C

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Q.9

If z in any complex number satisfying
$|z-1|=1$, then which of the following is correct?

(A) $\arg(z-1)=2\arg z$

(B) $2\arg(z)=\frac{2}{3}\arg\left(z^2-z\right)$

(C) $\arg(z-1)=\arg(z+1)$

(D) $\arg z=2\arg(z+1)$

Q.10

The value of $\operatorname{Arg}\left(i\ln\left(\frac{a-ib}{a+ib}\right)\right)$ where a, b are real numbers is

(A) 0 or π

(B) $\frac{\pi}{2}$

(C) $-\frac{\pi}{2}$ or $\frac{\pi}{2}$

(D) not defined

Q.11

Let $\mathrm{A}\left(z_1\right)$ and $\mathrm{B}\left(z_2\right)$ are such that $\angle \mathrm{AOB}=\theta$ ($'0^{\circ}$ being the orgin). If we define
$z_1 * z_2=\left|z_1\right|\left|z_2\right|\sin\theta$ then $z_1 * z_2$ is also equal to

(A) $\operatorname{Re}\left(z_1\bar{z}_2\right)$

(B) $\operatorname{Re}\left(\bar{z}_1 z_2\right)$

(C) $\operatorname{Im}\left(\bar{z}_1 z_2\right)$

(D) none of these

Q.12

If z_1, z_2 and z_3 are complex numbers such that
$\left|z_1\right|=\left|z_2\right|=\left|z_3\right|=\left|\frac{1}{z_1}+\frac{1}{z_2}+\frac{1}{z_3}\right|=1$
then $\left|z_1+z_2+z_3\right|$ is

(A) equal to 1

(B) less than 1

(C) greater than 3

(D) equal to 3

Q.13

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If $z = 2 + i$, then $(z - 1)(\bar{z} - 5) + (\bar{z} - 1)(z - 5)$ is equal to

(A) 2

(B) 7

(C) -1

(D) -4

Q.14

If $(1 - i)(2 - i)(3 - i) \ldots (1 - ni) = x - iy$, then $2.5.10 \ldots \left(1 + n^2\right) =$

(A) $x^2 - y^2$

(B) $x^2 + y^2$

(C) $2x^2 + 3y^2$

(D) $x^2 - 2y^2$

Q.15

If z is a complex number such that $z + |z| = 8 + 12i$, then the value of $\left|z^2\right|$ is equal to

(A) 228

(B) 144

(C) 121

(D) 169

Q.16

If $|z_1| = |z_2| = \ldots\ldots\ldots\ldots |z_n| = 1$, then the value of $|z_1 + z_2 + \ldots\ldots\ldots z_n| - \left|\frac{1}{z_1} + \frac{1}{z_2} + \ldots\ldots\ldots + \frac{1}{z_n}\right|$ is

(A) 0

(B) 1

(C) -1

(D) None

Q.17

Let Z and W be two complex numbers such that $|Z| \leq 1, |W| \leq 1$ and $|Z + iW| = |Z - i\bar{W}| = 2$. Then Z equals

(A) 1 or i

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(B) i or -i

(C) 1 or -1

(D) i or -1

Q.18

For what value of λ the sum of the squares of the roots of $x^2 + (2+\lambda)x - \frac{1}{2}(1+\lambda) = 0$ is minimum ?

(A) $3/2$

(B) 1

(C) $-5/2$

(D) $11/4$

Q.19

If the roots of the equation $ax^2 - bx + c = 0$ are α, then the roots of the equation $b^2cx^2 - ab^2x + a^3 = 0$ are

(A) $\frac{1}{\alpha^3+\alpha\beta}, \frac{1}{\beta^3+\alpha\beta}$

(B) $\frac{1}{\alpha^2+\alpha\beta}, \frac{1}{\beta^2+\alpha\beta}$

(C) $\frac{1}{\alpha^4+\alpha\beta}, \frac{1}{\beta^4+\alpha\beta}$

(D) None of these

Q.20

If x is real, the maximum value of $\frac{3x^2+9x+17}{3x^2+9x+7}$ is

(A) 41

(B) 43

(C) 58

(D) 85

Answers & Solutions

Q.1 Answer:

$24/5$

Solution:

As $|z| \geq 5, \quad \left|\frac{1}{z}\right| \leq \frac{1}{5}.$

Now $\left|z - \frac{1}{z}\right| \geq \left||z| - \left|\frac{1}{|z|}\right|\right| = |z| - \frac{1}{|z|} \geq 5 - \frac{1}{5} = \frac{24}{5}$

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The least value is attained when $z = 5$.

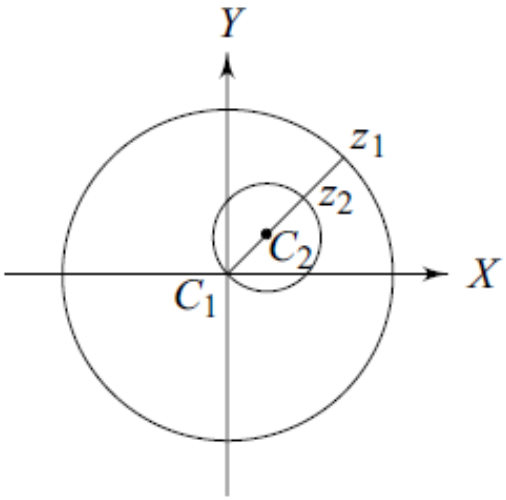
Q.2 Answer:

2

Solution:

$|z_1| = 12$ implies that z_1 lies on the circle with centre C_1 at the origin and radius 12 whereas $|z_2 - 3 - 4i| = 5$ implies z_2 lies on the circle with centre at $C_2(3 + 4i)$ and radius 5.

See Fig.



The quantity $|z_1 - z_2|$ will be least if z_1 and z_2 lie on the line joining C_1 and C_2
i.e. on the line $z = (3 + 4i)t$
In fact, when we take $z_2 = 6 + 8i$ and $z_1 = \frac{12}{5}(3 + 4i)$,
we obtain $|z_1 - z_2| = \left|\frac{12}{5}(3 + 4i) - (6 + 8i)\right| = \frac{2}{5}|3 + 4i| = 2$.

Q.3 Answer:

8

Solution:

$$(\bar{z} - 3i)(z + 3i) = 25$$
$$\Rightarrow \quad |z - 3i|^2 = 25 \text{ or } |z - 3i| = 5$$

Now,

$$|z| = |(z - 3i) + 3i|$$
$$\leq |z - 3i| + |3i| = 5 + 3 = 8$$

Q.4 Answer:

$\text{Re}\, z > 3$

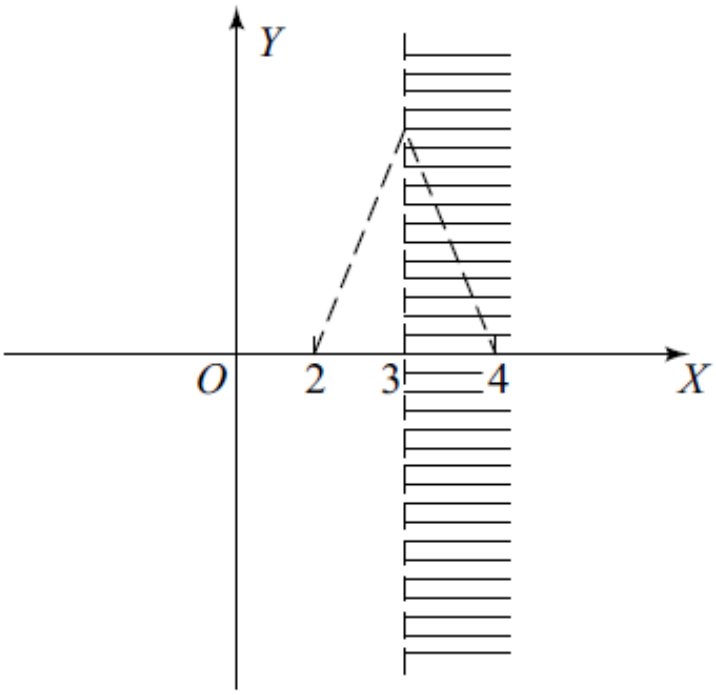
Solution:

If z satisfies $|z - 4| = |z - 2|$, then z lies on the perpendicular bisector of the segment joining $z = 2$ and $z = 4$.

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i.e., $|z-4| = |z-2| \Rightarrow \mathrm{Re}(z) = 3.$

As $z = 0$ **does not satisfy** $|z-4| < |z-2|$**, we get** $|z-4| < |z-2|$ **represents the region** $\mathrm{Re}(z) > 3$



Q.5 Answer:

$x = 0, y = 0$

Solution:

Given, $\begin{vmatrix} 6\mathrm{i} & -3\mathrm{i} & 1 \\ 4 & 3\mathrm{i} & -1 \\ 20 & 3 & \mathrm{i} \end{vmatrix} = \mathrm{x} + \mathrm{iy}$

Now, $\begin{vmatrix} 6i & -3\mathrm{i} & 1 \\ 4 & 3\mathrm{i} & -1 \\ 20 & 3 & \mathrm{i} \end{vmatrix}$

$= 6\mathrm{i}(-3+3) + 3\mathrm{i}(4\mathrm{i}+20) + 1(12-60\mathrm{i})$

$= 0 + 0\mathrm{i}$

Comparing with RHS, we get $x = 0, y = 0$

Q.6 Answer:

$\sqrt{7}$

Solution:

$z = x + iy$

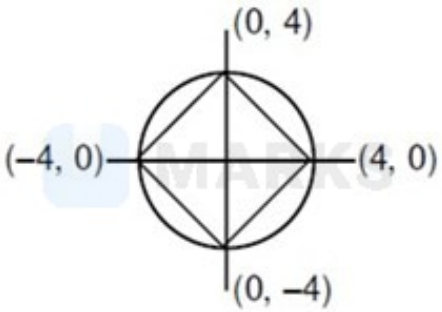
$x + y = 4$

Minimum value of $z = 2\sqrt{2}$

Maximum value of $z = 4$

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So, z can't be $\sqrt{7}$



Q.7 Answer:

0

Solution:

Let, $\frac{3}{z_1 - z_2} = \frac{5}{z_2 - z_3} = \frac{7}{z_3 - z_1} = k$

$\frac{9}{z_1 - z_2^2} = \frac{25}{z_2 - z_3^2} = \frac{49}{z_3 - z_1^2} = k^2$

$\Rightarrow \frac{9}{z_1 - z_2 \bar{z}_1 - \bar{z}_2} = k^2 \Rightarrow \frac{9}{z_1 - z_2} = k^2 z_1 - z_2$

similarly, $\frac{25}{z_2 - z_3} = k^2 z_2 - z_3$

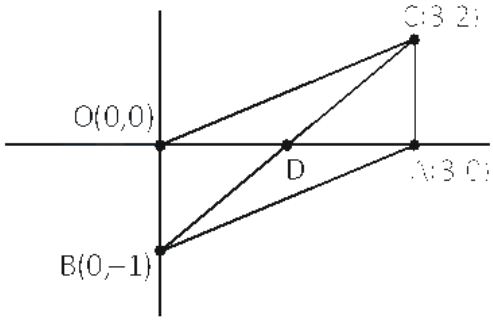
and $\frac{49}{z_3 - z_1} = k^2 \bar{z}_3 - \bar{z}_1$

$\Rightarrow \frac{9}{z_1 - z_2} + \frac{25}{z_2 - z_3} + \frac{49}{z_3 - z_1} = k^2 z_1 - z_2 + z_2 - z_3 + z_3 - z_1 = 0$

Q.8 Answer:

point of intersection of CB and AO

Solution:



The minimum value of $z + z$ - 3 occurs at points on the line segment joining O & A

Similarly, the minimum value of $z + i + z$ - 3 - 2i

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occurs at points on the line segment joining B & C.

$\Rightarrow$ The minimum value of $z + z - 3 + z + i + z - 3 - 2i$

occurs at point of intersection of OA & BC.

Q.9 Answer:

$\arg(z-1) = 2\arg z$

Solution:

Since $|z-1| = 1 \quad \therefore z - 1 = e^{i\theta}$**, where**

$\arg|z-1| = \theta$

$\therefore z = 1 + \cos\theta + i\sin\theta$

$= 2\cos\frac{\theta}{2}\left[\cos\frac{\theta}{2} + i\sin\frac{\theta}{2}\right]$

$= 2\cos\frac{\theta}{2}\cdot e^{i\theta/2} = 2\cos^2\frac{\theta}{2} + 2i\sin\frac{\theta}{2}\cos\frac{\theta}{2}$

Thus, $\arg(z-1) = 2\arg z$

Q.10 Answer:

0 or π

Solution:

$$\text{Arq}\left(i\ln\left(\frac{a-ib}{a+ib}\right)\right)$$

$$\text{Let atib } = z \quad \therefore z = |z|e^{i\theta}, \text{ where } |z| = \sqrt{a^2+b^2} \text{ and } \theta = \tan^{-1}\left(\frac{b}{a}\right)$$

$$\therefore a - ib = \bar{z} \quad \therefore \bar{z} = |z|e^{-i\theta}$$

$$\therefore \text{Arg}\left(i\ln\left(\frac{a-ib}{a+ib}\right)\right) = \text{Arg}\left(i\ln\left(\frac{\bar{z}}{z}\right)\right) = \text{Arg}\left(i\ln\left(\frac{|z|e^{-i\theta}}{|z|e^{-i\theta}}\right)\right)$$

$$= \text{Arg}\left(i\ln\left(e^{-Li\theta}\right)\right)$$

$$= \text{Arg}(i\times -2i\theta)$$

$$= \text{Arg}(2\theta)$$

and $\theta = \tan^{-1}\left(\frac{b}{a}\right) \quad \left(\because \frac{b}{a}\right.$ **is a real number)**

$\therefore \theta$ **is also a real number**

and $\arg(\theta) = 0$ **or** π **[because argument of a real number is 0 or** π**].**

Q.11 Answer:

$\text{Im}\left(\bar{z}_1 z_2\right)$

Solution:

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We have $\frac{z_2}{z_1}=\frac{|z_2|}{|z_1|}e^{i\theta}$

$\Rightarrow \frac{z_2}{z_1}|z_1|^2=|z_1||z_2|e^{i\theta}$

$\Rightarrow |z_1||z_2|\sin\theta=\text{Im}\left(\frac{z_2}{z_1}|z_1|^2\right)$

$\Rightarrow \text{Im}(z_2\bar{z}_1)$

Q.12 Answer:

equal to 1

Solution:

$|z_1|=|z_2|=|z_3|=1$

(given) Now, $|z_1|=1 \quad \Rightarrow |z_1|^2=1\Rightarrow z_1\bar{z}_1=1$

Similarly, $z_2\bar{z}_2=1, z_3\bar{z}_3=1$

Now, $\left|\frac{1}{z_1}+\frac{1}{z_2}+\frac{1}{z_3}\right|=1\Rightarrow|\bar{z}_1+\bar{z}_2+\bar{z}_3|=1$

$\Rightarrow \quad |\overline{z_1+z_2+z_3}|=1\Rightarrow|z_1+z_2+z_3|=1$

Q.13 Answer:

-4

Solution:

$(\mathrm{z}-1)(\bar{\mathrm{z}}-5)+(\bar{\mathrm{z}}-1)(\mathrm{z}-5)$

$=2\,\mathrm{Re}[(\mathrm{z}-1)(\bar{\mathrm{z}}-5)]$

$$[\because \mathrm{z}_1\bar{\mathrm{z}}_2+\mathrm{z}_2\bar{\mathrm{z}}_1=2\,\mathrm{Re}\,(\mathrm{z}_1\mathrm{z}_2)]$$

$$=2\,\mathrm{Re}[(1+\mathrm{i})(-3-\mathrm{i})]=2(-2)=-4$$

$$[\text{ Given } \mathrm{z}=2+\mathrm{i}]$$

Q.14 Answer:

x^2+y^2

Solution:

$(1+i)(1+2i)(1+3i)\ldots(1+ni)=x+iy$ **...(1)**

Taking conjugate, we get

$(1-i)(1-2i)(1-3i)\ldots(1-ni)=x-iy$ **...(2)**

Multiplying Eqs. (1) and (2), we get

$\left(1^2+1^2\right)\left(1^2+2^2\right)\ldots\left(1^2+n^2\right)=x^2+y^2$

So, $2.5.10\ldots\left(1+n^2\right)=x^2+y^2$

Q.15 Answer:

169

Solution:

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$z + |z| = 8 + 12i$

$\Rightarrow x + iy + \sqrt{x^2 + y^2} = 8 + 12i$

$\Rightarrow x + \sqrt{x^2 + y^2} = 8 \quad \ldots (\text{i}) \quad \& \quad y = 12 \quad \ldots (\text{ii})$

$(x = -5)$ **So,** $z = -5 + 12i$

$\Rightarrow |z| = \sqrt{25 + 144} = 13 \quad \Rightarrow |z^2| = |z|^2 = 169$

Q.16 Answer:

0

Solution:

$z_1\bar{z}_1 = z_2\bar{z}_2 = \ldots\ldots = z_n\bar{z}_n = 1$

$\Rightarrow \bar{z}_1 = \frac{1}{z_1}, \bar{z}_2 = \frac{1}{z_2}, \bar{z}_3 = \frac{1}{z_3}, \ldots\ldots, \bar{z}_n = \frac{1}{z_n}$

$\therefore |z_1 + z_2 + \ldots\ldots + z_n| - \left|\frac{1}{z_1} + \frac{1}{z_2} + \ldots + \frac{1}{z_n}\right|$

$= |z_1 + z_2 + \ldots + z_n| - |\bar{z}_1 + \bar{z}_2 + \ldots\ldots + \bar{z}_n| = 0$

Q.17 Answer:

1 or -1

Solution:

We have $2 = |z + i\omega| \leq |z| + |\omega|$**...(1)**

$\therefore \quad |z| + |\omega| \geq 2$

But given that $|z| \leq 1$ **and** $|\omega| \leq 1 \quad \ldots$ **(i**

(ii) $\Rightarrow |z| + |\omega| \leq 2$

From (1) and $(2)|z| = |\omega| = 1$

Also $|z + i\omega| = |z - i\bar{\omega}| \Rightarrow \quad |z + i\omega|^2 = |z - i\bar{\omega}|^2$

$\Rightarrow \quad (z + i\omega)(\bar{z} - i\bar{\omega}) = (\bar{z} + i\omega)(z - i\bar{\omega})$

$\Rightarrow \quad z\bar{z} + i\omega\bar{z} - iz\bar{\omega} + \omega\bar{\omega} = z\bar{z} - i\overline{z\omega} + i\omega z + \omega\bar{\omega}$

$\Rightarrow \quad \omega\bar{z} - \bar{\omega}z + \bar{\omega}\bar{z} - \omega z = 0$

$\Rightarrow (\omega + \bar{\omega})(\bar{z} - z) = 0$

$\Rightarrow z = \bar{z}$ **or** $\omega = -\bar{\omega} \Rightarrow \text{Im}(z) = 0 \Rightarrow \text{Re}(\omega) = 0$

Also $|z| = 1, |\omega| = 1 \Rightarrow z = 1$ **or** -1 **and** $\omega = i$

or $-i$

Q.18 Answer:

$-5/2$

Solution:

Given equation is

$x^2 + (2 + \lambda)x - \frac{1}{2}(1 + \lambda) = 0$

So $\alpha + \beta = -(2 + \lambda) = 0$ **and** $\alpha\beta = -\left(\frac{1+\lambda}{2}\right)$

Now, $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

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$\Rightarrow \alpha^2+\beta^2=[-(2+\lambda)]^2+2\frac{(1+\lambda)}{2}$

$\Rightarrow \alpha^2+\beta^2=\lambda^2+4+4\lambda+1+\lambda$

$=\lambda^2+5\lambda+5$

Which is for $\lambda=1/2$

Q.19 Answer:

$\frac{1}{\alpha^2+\alpha\beta}, \frac{1}{\beta^2+\alpha\beta}$

Solution:

Multiplying the second equation by $\frac{c}{a^3}$

we get $\frac{b^2c^2}{a^3}x^2-\frac{b^2c}{a^2}x+c=0$

$\Rightarrow \quad a\left(\frac{bc}{a^2}x\right)^2-b\left(\frac{bc}{a^2}\right)x+c=0$

$\Rightarrow \frac{bc}{a^2}x=\alpha,\beta$

$\Rightarrow \quad (\alpha+\beta)\alpha\beta x=\alpha,\beta$

$\Rightarrow x=\frac{1}{(\alpha+\beta)\alpha}, \frac{1}{(\alpha+\beta)\beta}$

Q.20 Answer:

41

Solution:

$y=\frac{3x^2+9x+17}{3x^2+9x+7}$

$3x^2(y-1)+9x(y-1)+7y-17=0$

$D\geq 0 \because$ **x is real**

$81(y-1)^2-4\times 3(y-1)(7y-17)\geq 0$

$\Rightarrow \quad (y-1)(y-41)\leq 0 \Rightarrow 1\leq y\leq 41$

$\therefore$ **Max value of** y **is 41**

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